

Superstore Sales Analysis

End-to-End Business Intelligence Project

Tools Used: Excel, Power Query, Pivot Tables, Pivot Charts, Interactive Dashboard

1. Project Overview

Business Problem

The company aims to understand its sales performance, profitability, customer behavior, and operational efficiency to support data-driven decision-making. The objective of this analysis is to identify key business trends, uncover profit drivers, detect underperforming products, and provide actionable recommendations for improving business performance.

2. Dataset Description

The dataset contains historical sales transactions from a global retail company between 2012 and 2015.

Dataset Includes

- Orders
 - Customers
 - Products
 - Categories
 - Markets
 - Regions
 - Shipping Information
 - Sales
 - Profit
 - Quantity
 - Discount
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3. Data Cleaning

Before starting the analysis, the dataset was cleaned using Power Query.

Data Cleaning Steps

- Removed unnecessary metadata rows.
- Verified data types for all columns.
- Checked for duplicate records.
- Validated Order Date and Ship Date consistency.
- Created a new Delivery Days column.
- Handled missing Postal Code values.
- Verified that there were no invalid delivery durations.
- Reviewed data quality before analysis.

4. Executive KPIs

KPI	Value
Revenue	\$12.64M
Profit	\$1.47M
Profit Margin	12%
Orders	25,722
Customers	17,413
Average Order Value	\$491.49

5. Sales Performance Analysis

Key Findings

- Revenue and profit increased consistently across the four-year period.
- 2015 achieved the highest revenue and profit.
- Quarter 4 generated the highest sales and profitability every year.
- November and December were consistently among the strongest-performing months.
- The business demonstrates a clear seasonal sales pattern.

Business Insight

Sales continue to grow year after year, while Quarter 4 represents the company's strongest selling season. This indicates an opportunity to maximize revenue through targeted campaigns and inventory planning before year-end.

6. Product Analysis

Key Findings

- Copiers generated the highest profit.
- Phones ranked among the strongest-performing products.
- Tables consistently generated losses despite generating revenue.
- Several office supply products sold high quantities but delivered relatively low profitability.

Business Insight

High sales volume does not necessarily translate into high profitability. Product profitability should be considered alongside revenue when making pricing and inventory decisions.

Recommendation

Review pricing, discount strategy, and procurement costs for loss-making products such as Tables.

7. Customer Analysis

Key Findings

- Consumer is the highest-performing customer segment.
- Top 10 customers contribute only:
 - 1% of total revenue
 - 3% of total profit
- High-spending customers are not always the most profitable customers.

Business Insight

The business is not dependent on a small number of customers, reducing customer concentration risk. Customer profitability varies significantly, suggesting that discount strategies and product mix influence overall profitability.

8. Market & Geographic Analysis

Key Findings

- Asia Pacific generated the highest revenue.
- Europe generated the highest profit.
- Western Europe achieved the highest revenue, profit, and profit margin.
- Top 10 regions contributed:
 - 74% of total revenue
 - 86% of total profit

Business Insight

Revenue is concentrated across a limited number of regions, while Western Europe consistently outperforms all other regions in profitability and operational efficiency.

Recommendation

Investigate the business practices contributing to Western Europe's success and evaluate opportunities to replicate them in other markets.

9. Shipping Analysis

Key Findings

- Standard Class accounted for approximately 60% of both orders and revenue.
- Average delivery times aligned with each shipping mode:
 - Same Day → 0 days
 - First Class → 2 days
 - Second Class → 3 days
 - Standard Class → 5 days

Business Insight

Shipping performance is operationally consistent, with Standard Class serving as the primary shipping method without negatively impacting business performance.

10. Discount Analysis

Key Findings

- The highest sales occurred without discounts.
- Discounts did not consistently increase sales volume.
- Discounts of 20% or more were associated with negative profitability.
- All product categories experienced declining profitability as discount levels increased.

Business Insight

Heavy discounting significantly reduces profitability without guaranteeing higher sales performance.

Recommendation

Reevaluate the current discount strategy and limit high-discount promotions unless they serve a specific business objective.

11. Executive Summary

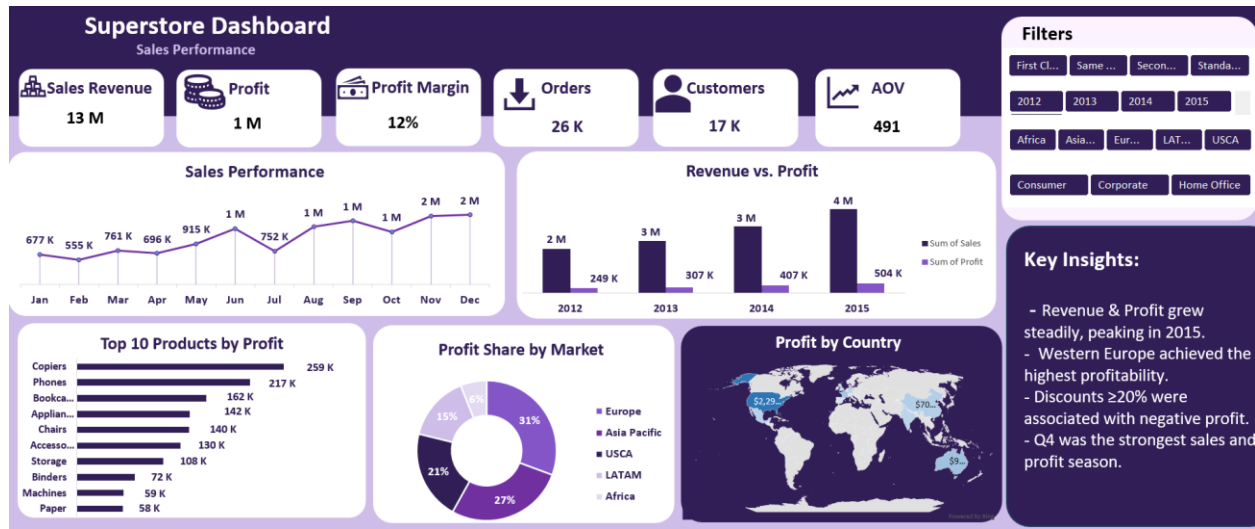
Key Business Insights

- Revenue and profit increased steadily, reaching their highest levels in 2015.
 - Quarter 4 consistently delivered the strongest sales performance.
 - Western Europe is the company's most profitable region.
 - Discounts above 20% are associated with negative profitability.
 - Product profitability should be prioritized over revenue alone when making business decisions.
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12. Business Recommendations

- **Review the discount strategy to improve overall profitability.**
 - **Prioritize investment in high-margin products such as Copiers and Phones.**
 - **Investigate the causes of persistent losses in Tables.**
 - **Expand successful operational strategies from Western Europe to other regions.**
 - **Increase inventory and marketing activities before Quarter 4 to capitalize on seasonal demand.**
 - **Monitor customer profitability in addition to customer spending.**
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13. Dashboard



14. Project Skills Demonstrated

- Data Cleaning using Power Query
- Data Validation
- KPI Development
- Pivot Tables & Pivot Charts
- Business Performance Analysis
- Customer Segmentation
- Product Performance Analysis
- Geographic Analysis
- Discount & Profitability Analysis
- Interactive Dashboard Design
- Business Insight Generation
- Executive Reporting